

Section 2 – Potential Flow

Possible assumptions

Steady:

$$\frac{\partial(\quad)}{\partial t} \approx 0 \quad (2.1)$$

Incompressible:

$$\frac{\partial \rho}{\partial(\quad)} \approx 0 \quad \text{or} \quad \rho \approx \text{constant} \quad (2.2)$$

Inviscid:

$$F_{visc} \approx 0 \quad (2.3)$$

No body forces:

$$\rho g_x = \rho g_y = \rho g_z \approx 0 \quad (2.4)$$

Inviscid Flow (TSec. 3.1)

Focus for a while on the differential form of the momentum eq. (1.20)

$$\frac{\partial \vec{V}}{\partial t} + (\vec{V} \cdot \nabla) \vec{V} = -\frac{1}{\rho} \nabla p + \vec{f} + \frac{1}{\rho} (\vec{F}_{visc} + \vec{F}_{ext})$$

Assume the flow is inviscid (negligible viscous forces), and no other external forces.

$$\frac{\partial \vec{V}}{\partial t} + (\vec{V} \cdot \nabla) \vec{V} = -\frac{1}{\rho} \nabla p + \vec{f} \quad (\text{TEq. (3.2)}) \quad (2.5)$$

When is the flow inviscid?

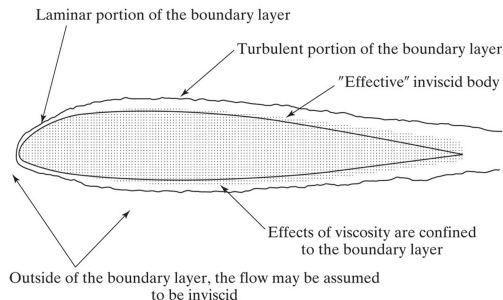


Figure 2.1: Fig. 2.14 from AE-BC

- ▶ Outside the boundary layer and wake - **potential flow** (inviscid, irrotational & steady - more on this soon)
- ▶ Boundary layer and wake - **unsteady, viscous & rotational**

NOTE: Remember inviscid only means that viscous forces are negligible compared to other forces in the system. If viscosity is zero then it is not a fluid.

When is the flow inviscid?

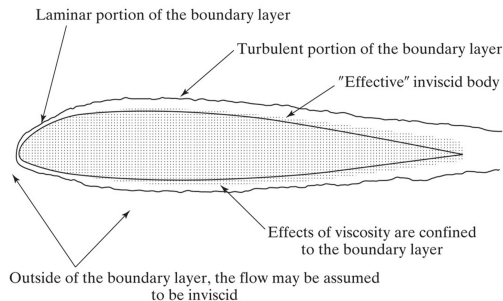


Figure 2.2: Fig. 2.14 from AE-BC

Note,

$$\text{Reynolds number} = \frac{\text{Inertial forces}}{\text{Viscous forces}} \quad (2.6)$$

Therefore, in the case of inviscid flow,

$$\text{Reynolds number} \rightarrow \infty \quad (2.7)$$

Pathlines - AE 201 revision

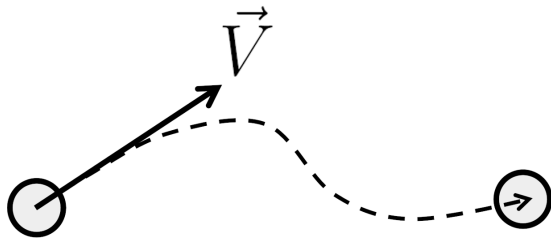


Figure 2.3: A pathline followed by a fluid element. Also shown is the velocity vector at the initial starting point.

Pathlines are **paths followed by individual fluid elements**. We usually obtain this by integrating in time. We start at a point and integrate forward.

Streamlines - AE 201 revision

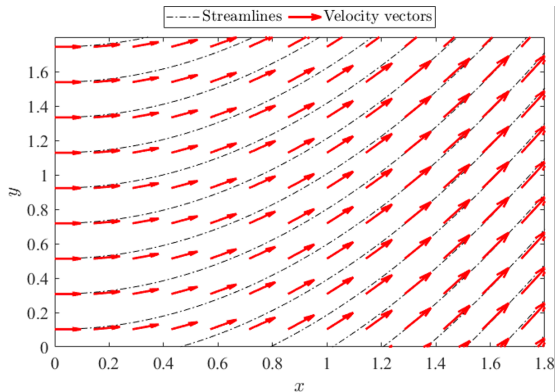


Figure 2.4: Streamlines in a flow. The black lines are the streamlines while the red vectors are the velocity vectors which are tangent everywhere to the streamlines.

Streamline: - is a curve whose tangent at every point is the **direction** of the velocity vector $\vec{V}$ at that point

Streamlines - AE 201 revision

Let $\vec{ds}$ be the streamline in a flow and let $\vec{V}$ be the velocity vector.

$$\vec{ds} = dx \hat{i} + dy \hat{j} + dz \hat{k}$$

$$\vec{V} = u \hat{i} + v \hat{j} + w \hat{k}$$

For a vector to be parallel to another vector the **curl** has to be **zero**.

$$\vec{ds} \times \vec{V} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ dx & dy & dz \\ u & v & w \end{vmatrix} = 0 \quad (2.8)$$

$$= (w \, dy - v \, dz) \hat{i} + (u \, dz - w \, dx) \hat{j} + (v \, dx - u \, dy) \hat{k} = 0 \quad (2.9)$$

Therefore, the equation of a streamline:

$$w \, dy - v \, dz = u \, dz - w \, dx = v \, dx - u \, dy = 0 \quad (2.10)$$

Streakline - AE 201 revision

A streakline is the locus of points formed by fluid elements **that have passed through a certain point in space**. In general, fluid elements that pass through a certain point will follow different pathlines and so end up at different points in the flow. The smoke trace that you observe in the laboratory is an example of a streakline.

What happens when the flow is steady?

The pathlines, streakline and streamline **all become the same line**.

Irrotational & rotational flow (TSec. 3.6)

In 3-D the angular velocity:

$$\vec{\omega} = \frac{1}{2} \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ u & v & w \end{vmatrix} = \frac{1}{2} \vec{\nabla} \times \vec{V} \quad (2.11)$$

In cylindrical coordinates the angular velocity:

$$\vec{\omega} = \frac{1}{2r} \begin{vmatrix} \hat{e}_r & r\hat{e}_\theta & \hat{e}_z \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial z} \\ v_r & rv_\theta & v_z \end{vmatrix} = \frac{1}{2} \vec{\nabla} \times \vec{V} \quad (2.12)$$

Note, here $\times$ is the “curl” operator. Also, $\vec{\omega}$ is a vector.

Irrotational and rotational flow

A fluid element in an irrotational flow only **translates** and **deforms**. It does not **rotate**. Mathematically, a flow is irrotational when

$$\vec{\nabla} \times \vec{V} = 2\vec{\omega} = 0 \quad (\text{TEq. (3.17)}) \quad (2.13)$$

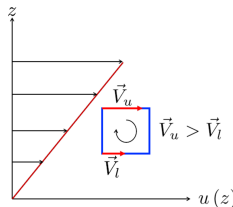


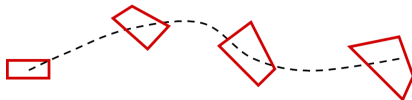
Figure 2.5: Fluid element in a flow field rotating due to the velocity gradient it is subject to

Viscous flows **ARE** rotational. **Some** inviscid flows are rotational.

Irrotational and rotational flow

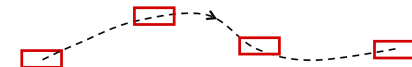


Rotational flow, but no distortion or change in volume

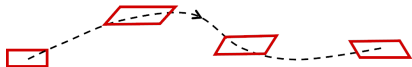


Rotational flow, with distortion and change in volume

Figure 2.6: Rotational flow



Irrotational flow with no distortion or change in volume



Irrotational flow, with distortion and change in volume

Figure 2.7: Irrotational flow

Circulation (TSec. 3.5)

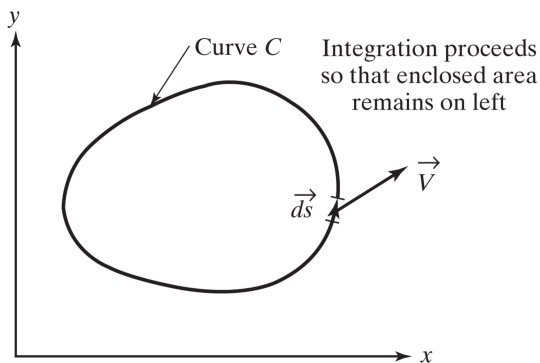


Figure 2.8: Fig. 3.4 from AE-BC

$$\text{Circulation } \Gamma = - \oint_C \vec{V} \cdot \vec{ds} \quad (\text{TEq. (3.14)}) \quad (2.14)$$

The $\oint_c$ here indicates integral over a **closed** surface. Also, circulation Γ here is defined as being **positive** in a clockwise sense (Ensures positive lift!)

Circulation

Using Stokes theorem,

$$\Gamma = - \oint_C \vec{V} \cdot d\vec{s} = - \iint_A (\vec{\nabla} \times \vec{V}) \cdot \hat{n} dA \quad (\text{TEq. (3.16)}) \quad (2.15)$$

Therefore, if the flow within the contour of integration is irrotational everywhere, $\Gamma = 0$ as $\vec{\nabla} \times \vec{V} = 2\vec{\omega} = 0$

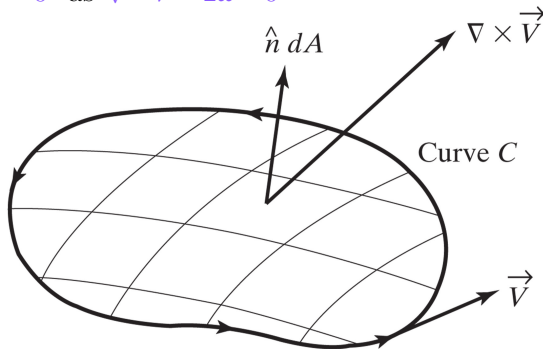


Figure 2.9: Fig. 3.6 from AE-BC

Circulation

Flow having circulation does not mean fluid particles are going around in circles! Typically the streamlines need to have some sort of curvature for finite circulation to exist. In other words the flow has to turn in some manner as in flow over an airfoil

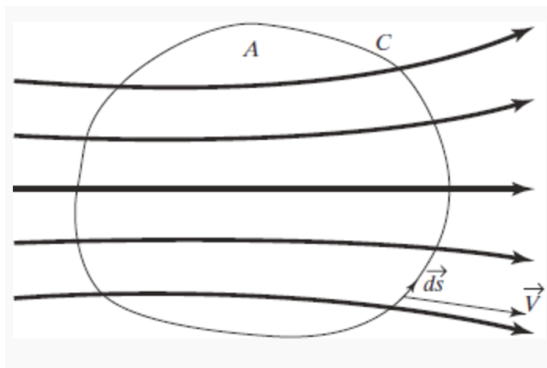


Figure 2.10: from AE-BC

When is flow incompressible?

In general, compressibility effects are insignificant at low speeds and the density can be assumed to be constant.

$$\frac{\rho_{t1}}{\rho} = \left(1 + \frac{\gamma - 1}{2} M_\infty^2\right)^{1/(\gamma-1)} \quad (\text{TEq. (8.36)}) \quad (2.16)$$

Eq. (2.16) will be derived later in the course. Here, ρ_{t1} is the stagnation density and γ is the ratio of specific heats (see (TEq. (8,15))).

Note for air ($\gamma = 1.4$) at $M_\infty = 0.3$

$$\frac{\rho_{t1}}{\rho} = 1.0456 \quad \text{i.e., 5\% variation in } \rho \quad (2.17)$$

Therefore, as a rule of thumb, we assume the flow to be incompressible when $M_\infty \leq 0.3$.

Note, nothing magical happens between $M_\infty = 0.29$ and $M_\infty = 0.31$.

Potential flow assumptions

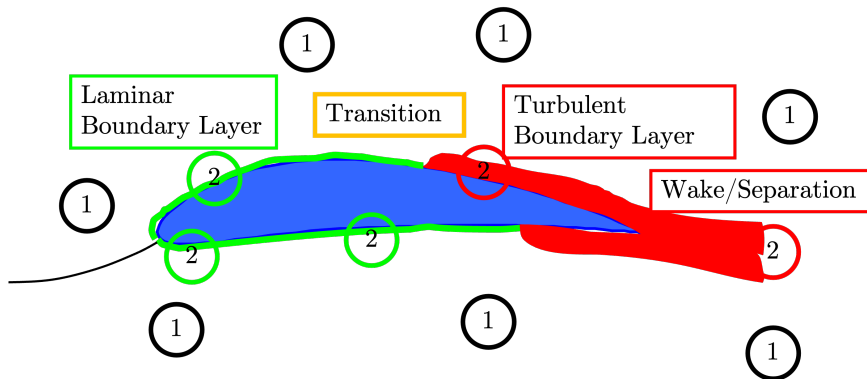
Under certain conditions, parts of the flow can be assumed to be

1. **Steady** i.e., $\frac{\partial(\quad)}{\partial t} \approx 0$
2. **Incompressible** i.e., $M_\infty \leq 0.3$ or $\rho \approx \text{constant}$
3. **Irrotational** (implies inviscid) i.e., $\nabla \times \vec{V} = 0$

Collectively, when these assumptions are imposed on a flow field we consider the flow to be “ideal” or “potential” flow. I call these the 3.5 assumptions of potential flows.

Initially we shall focus on 2-D potential flows (airfoils) and then extend it to 3-D flows (wings). In aerodynamics, when we consider flow over airfoils, wings etc., everything is at the same altitude so we neglect gravitational forces (the most common form of body force). But this is not an ideal or potential flow assumption. So to be rigorous we are going to consider potential flows with no body force effects.

Potential flow



- ▶ Region 1 - Outside the boundary layer and wake at low speeds the flow can be considered to be potential.
- ▶ Region 2 - Within the boundary layer and wake, even at low incompressible speeds, the flow is rotational (inviscid) and unsteady.

Potential function (TSec. 3.8)

2-D mass conservation (Eq. (1.12)):

$$\frac{\partial \rho}{\partial t} + \vec{\nabla} \cdot (\rho \vec{V}) = 0 \quad (2.18)$$

$$\text{Steady: } \rightarrow \frac{\partial \rho}{\partial t} \approx 0 \quad (2.19)$$

$$\text{Incompressible: } \rightarrow \rho \approx \text{constant} \quad (2.20)$$

Therefore, 2-D continuity equation for **steady, incompressible** flow:

$$\vec{\nabla} \cdot \vec{V} = 0 \rightarrow \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \quad (2.21)$$

Potential fn.

Consider an irrotational flow, then:

$$\vec{\nabla} \times \vec{V} = 2\vec{\omega} = 0 \quad (2.22)$$

Vector calculus relationship:

$$\vec{\nabla} \times (\vec{\nabla} \phi) = 0, \text{ where } \phi \text{ is a scalar} \quad (2.23)$$

$$\text{Therefore, } \vec{V} = \vec{\nabla} \phi \quad (2.24)$$

That is IF the flow is irrotational then, the velocity vector $\vec{V}$ can be expressed as the gradient of a scalar ϕ . This scalar is called the velocity potential function or simply the velocity potential.

Potential fn.

$$\text{Therefore, } \vec{V} = \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \phi \quad (\text{TEq. (3.19)}) \quad (2.25)$$

$$\text{Therefore, } u = \frac{\partial \phi}{\partial x}, \quad v = \frac{\partial \phi}{\partial y}, \quad w = \frac{\partial \phi}{\partial z} \quad (2.26)$$

Chain rule (2-D):

$$d\phi = \frac{\partial \phi}{\partial x} dx + \frac{\partial \phi}{\partial y} dy \quad (2.27)$$

$$= u \, dx + v \, dy \quad (2.28)$$

Lines of equal potential or equipotential lines: $\rightarrow d\phi = 0$ Therefore, one way to describe equipotential lines is

$$\left. \frac{dy}{dx} \right|_{\phi=c} = -\frac{u}{v} \quad (2.29)$$

2-D Cylindrical coordinates

$$\text{Velocities: } v_r(r, \theta) \quad v_\theta(r, \theta) \quad v_z = 0 \quad (2.30)$$

$$\text{Potential function.: } v_r = \frac{\partial \phi}{\partial r} \quad v_\theta = \frac{1}{r} \frac{\partial \phi}{\partial \theta} \quad (2.31)$$

Stream function (TSec. 3.9)

Eq. of a streamline from eq. (2.10) for 2-d flow is given by,

$$\frac{dy}{dx} = \frac{v}{u} \rightarrow u \, dy - v \, dx = 0 = d\psi \quad (2.32)$$

$$\text{Therefore, } d\psi = 0 \rightarrow \psi = \text{constant} \quad (2.33)$$

Therefore, $\psi(x, y) = c$ is a streamline where ψ is called the **stream function** and c is any constant. Note, two important facts:

- ▶ The flow field doesn't have to be potential to define a stream function.
- ▶ The velocity $\vec{V}$ is tangent at every point on the line $\psi = c$ hence if we consider two streamlines $\psi = c_1$ and $\psi = c_2$ the mass flow between these two lines should be conserved.

Stream function

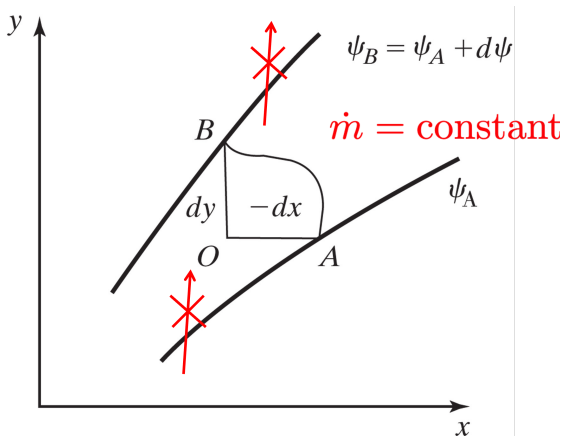


Figure 2.11: Modified Fig 3.7 from AE-BC

Stream function

Use Chain rule:

$$d\psi = \frac{\partial\psi}{\partial x}dx + \frac{\partial\psi}{\partial y}dy \quad (2.34)$$

Therefore, from eq. (2.32)

$$u = \frac{\partial\psi}{\partial y}, \quad v = -\frac{\partial\psi}{\partial x} \quad (2.35)$$

Hence the more formal way of writing the equation of a streamline is

$$\left. \frac{dy}{dx} \right|_{\psi=c} = \frac{v}{u} \quad (2.36)$$

Stream fn. for steady & incompressible flow

From eq. (2.35)

$$u = \frac{\partial \psi}{\partial y}, \quad v = -\frac{\partial \psi}{\partial x}$$

Therefore, $\nabla \cdot \vec{V} = 0$ (eq. (2.21)), which is the steady, incompressible continuity equation, becomes

$$\frac{\partial^2 \psi}{\partial x \partial y} - \frac{\partial^2 \psi}{\partial y \partial x} = 0 \quad (2.37)$$

As long as ψ is a reasonable function then $\frac{\partial^2 \psi}{\partial x \partial y} = \frac{\partial^2 \psi}{\partial y \partial x}$.

Hence, for steady, incompressible flow ψ satisfies the continuity equation

2-D Cylindrical coordinates

$$\text{Velocities: } v_r(r, \theta) \quad v_\theta(r, \theta) \quad v_z = 0 \quad (2.38)$$

$$\text{Stream function: } v_r = \frac{1}{r} \frac{\partial \psi}{\partial \theta} \quad v_\theta = -\frac{\partial \psi}{\partial r} \quad (2.39)$$

Equipotential lines & streamlines (TSec. 3.10)

Lines of constant ϕ (equipotential lines)

$$d\phi = 0 \rightarrow u \, dx = -v \, dy \rightarrow \left. \frac{dy}{dx} \right|_{\phi=\text{const}} = -\frac{u}{v} \quad (2.40)$$

Lines of constant ψ (streamlines)

$$d\psi = 0 \rightarrow u \, dy = v \, dx \rightarrow \left. \frac{dy}{dx} \right|_{\psi=\text{const}} = \frac{v}{u} \quad (2.41)$$

Therefore,

$$\left. \frac{dy}{dx} \right|_{\phi=\text{const}} \times \left. \frac{dy}{dx} \right|_{\psi=\text{const}} = -1 \quad (2.42)$$

Therefore, equipotential lines are everywhere **perpendicular** to streamlines. Hence, if you know one you can calculate the other to within a constant.

Equipotential lines and streamlines

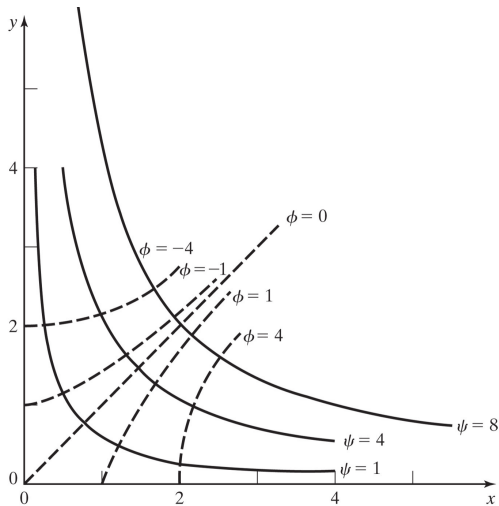


Figure 2.12: Fig 3.8 from AE-BC

Laplace eq. (TSec. 3.8 & 3.9)

Eq. (2.21) - Continuity equation for steady and incompressible flow:

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 = \vec{\nabla} \cdot \vec{V}$$

Now if the flow is irrotational $\rightarrow \vec{V} = \vec{\nabla}\phi$. Substituting in eq. (2.21) we get,

$$\vec{\nabla} \cdot \vec{\nabla}\phi = \nabla^2\phi = 0 = \frac{\partial^2\phi}{\partial x^2} + \frac{\partial^2\phi}{\partial y^2} \quad (2.43)$$

Further, as the flow is irrotational, $\vec{\nabla} \times \vec{V} = 2\vec{\omega} = \mathbf{0}$. Considering 2-D flow,

$$\vec{\nabla} \times \vec{V} = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} = 0 \quad (2.44)$$

$$\text{From eq. (2.35)} \quad u = \frac{\partial\psi}{\partial y}, \quad v = -\frac{\partial\psi}{\partial x}$$

Laplace eq.

Substituting eq. (2.35) into eq. (2.44) we get,

$$\nabla^2 \psi = 0 = \frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} \quad (2.45)$$

Hence, when the flow is steady, incompressible, and irrotational (inviscid), both the potential function ϕ and the streamfunction ψ satisfies Laplace's equation (Eq. (2.43) and (2.45) respectively.)

Remember, irrotationality implies that the flow is **inviscid**

Laplace eq.

To summarize if the flow is **steady, incompressible & irrotational (inviscid)** the stream function ψ (**2-D**) and the potential function ϕ both satisfy Laplace's equation

$$\nabla^2 \psi = \nabla^2 \phi = 0$$

Such a flow field is referred to as a **potential flow field** i.e., **steady, incompressible & irrotational (inviscid)** . This is also called ideal flow as such a flow does not exist in nature.

It is a purely mathematical construct

Why is this a big deal?

To recap we have shown that if the flow is steady, incompressible & irrotational (inviscid) the continuity eq. reduces to the Laplace Eq. (Eq. 2.43 & 2.45) $\rightarrow \nabla^2 \phi = \nabla^2 \psi = 0$.

- ▶ 1 equation, 1 unknown $\rightarrow$ Euler's eq. 4 eqns. 4 unknowns
- ▶ **Linear** equation $\rightarrow$ can use **principle of superposition** $\rightarrow$ if $\phi_1, \phi_2, \dots, \phi_n$ are all solutions to a linear equation then the linear combination of any of the solutions are also a solution. e.g., $\phi_i + 1/3\phi_j + 5\phi_k$ is also a solution. We shall exploit this. We shall find some basic solutions to the Laplace equation and then find linear combinations of them to yield other solutions.

Bernoulli's Eq. (TSec. 3.2)

Lets start with the steady, two-dimensional, inviscid momentum equations without body forces (eq. (2.5)) and the continuity equation (eq. (1.12))

$$\text{Continuity: } \frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} = \vec{\nabla} \cdot (\rho \vec{V}) = 0 \quad (2.46)$$

$$\text{x - momentum: } \vec{\nabla} \cdot (\rho \vec{V} u) = -\frac{\partial p}{\partial x} \quad (2.47)$$

$$\text{y - momentum: } \vec{\nabla} \cdot (\rho \vec{V} v) = -\frac{\partial p}{\partial y} \quad (2.48)$$

Governing eqn.

Also observe the following vector identity,

$$\vec{\nabla} \cdot (s\vec{F}) = s(\vec{\nabla} \cdot \vec{F}) + \vec{F} \cdot (\vec{\nabla} s) \quad (2.49)$$

where, s is any scalar and $\vec{F}$ any vector. Apply this identity to the momentum equations and then use the fact that from the steady continuity $\rightarrow \vec{\nabla} \cdot (\rho \vec{V}) = 0$. Therefore, the momentum equations become:

$$\text{x - momentum: } \rho \vec{V} \cdot (\vec{\nabla} u) = -\frac{\partial p}{\partial x} \quad (2.50)$$

$$\text{y - momentum: } \rho \vec{V} \cdot (\vec{\nabla} v) = -\frac{\partial p}{\partial y} \quad (2.51)$$

Therefore the density can be taken out of the gradient operator without having to assume incompressibility. It's been shown for the steady equations, also can be shown for the unsteady equations.

Bernoulli's eq.

Eq. (2.50) expanded out is,

$$u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = -\frac{1}{\rho} \frac{\partial p}{\partial x} \quad (2.52)$$

Multiply eq. (2.52) by dx :

$$u \frac{\partial u}{\partial x} dx + v \frac{\partial u}{\partial y} dx = -\frac{1}{\rho} \frac{\partial p}{\partial x} dx \quad (2.53)$$

Consider **irrotational** flow, i.e., $\vec{\nabla} \times \vec{V} = 0$

$$\vec{\nabla} \times \vec{V} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ u & v & w \end{vmatrix} = 0 \quad 2-D \rightarrow \frac{\partial v}{\partial x} = \frac{\partial u}{\partial y} \quad (2.54)$$

Bernoulli's eq.

Substituting eq. (2.54) into eq. (2.53) we get,

$$u \frac{\partial u}{\partial x} dx + v \frac{\partial v}{\partial x} dx = -\frac{1}{\rho} \frac{\partial p}{\partial x} dx \quad (2.55)$$

Similarly, from the y -momentum we get,

$$u \frac{\partial u}{\partial y} dy + v \frac{\partial v}{\partial y} dy = -\frac{1}{\rho} \frac{\partial p}{\partial y} dy \quad (2.56)$$

Adding eq. (2.55) and 2.56,

$$u \left(\frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy \right) + v \left(\frac{\partial v}{\partial x} dx + \frac{\partial v}{\partial y} dy \right) = -\frac{1}{\rho} \left(\frac{\partial p}{\partial x} dx + \frac{\partial p}{\partial y} dy \right) \quad (2.57)$$

Bernoulli's eq.

Remember, by chain rule

$$du = \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy \quad (2.58)$$

Therefore, eq. (2.57) becomes,

$$u \, du + v \, dv = -\frac{1}{\rho} dp \quad (2.59)$$

$$U = |\vec{V}| = \sqrt{u^2 + v^2} \quad (2.60)$$

$$U^2 = |\vec{V}|^2 = u^2 + v^2 \quad (2.61)$$

$$d(U^2) = d(u^2 + v^2) = 2u \, du + 2v \, dv \quad (2.62)$$

$$d\left[\frac{U^2}{2}\right] = u \, du + v \, dv \quad (2.63)$$

Compressible Bernoulli's eq.

Comparing eq. (2.63) in eq. (2.59) we get,

$$d\left[\frac{U^2}{2}\right] = -\frac{dp}{\rho} \rightarrow \underline{\text{Compressible Bernoulli's eq.}} \quad (2.64)$$

Assumptions:

- ▶ steady
- ▶ inviscid
- ▶ irrotational

Also in this form, it is assumed that there are no body forces and can be included as in your text.

To solve we integrate Eq. 2.64

Incompressible Bernoulli's eq.

If the flow is **incompressible**, $\rho \approx \text{constant}$ we can integrate eq. 2.64

$$\int \frac{dp}{\rho} = - \int d\left[\frac{U^2}{2}\right] \quad (2.65)$$

$$\frac{p}{\rho} + \frac{U^2}{2} = \text{const} \quad (2.66)$$

Or it is rewritten more commonly in the form,

$$p + \frac{1}{2}\rho U^2 = \text{const} \quad (2.67)$$

Also in this form it is assumed that there are no body forces and can be included as in your text.

Incompressible Bernoulli's eq.

$$p + \frac{1}{2}\rho U^2 = \text{const}$$

Assumptions:

- ▶ steady
- ▶ inviscid
- ▶ irrotational
- ▶ incompressible

Remember, these assumptions collectively are the **potential** flow assumptions.

$$C_p = \frac{p - p_\infty}{0.5\rho_\infty U_\infty^2} \quad (2.68)$$

When the flow is potential i.e., steady, incompressible, irrotational (inviscid), and when we ignore body forces, we can apply Bernoulli's equation between a point in the flow and a point in the free-stream.

$$p + \frac{1}{2}\rho_\infty U^2 = p_\infty + \frac{1}{2}\rho_\infty U_\infty^2 \quad (2.69)$$

$$p - p_\infty = \frac{1}{2}\rho_\infty U_\infty^2 \left[1 - \left(\frac{U}{U_\infty} \right)^2 \right] \quad (2.70)$$

Therefore, when the flow is steady, incompressible & irrotational (inviscid) (together these are the potential flow assumptions) and when we ignore body forces

$$C_p = 1 - \left(\frac{U}{U_\infty} \right)^2 \quad (2.71)$$

$$\text{At a stagnation point: } \rightarrow U = 0 \rightarrow C_p = 1 \quad (2.72)$$

$$\text{At a point where } \rightarrow U = U_\infty \rightarrow p = p_\infty \rightarrow C_p = 0 \quad (2.73)$$

$$\text{If } p > p_\infty \rightarrow U < U_\infty \rightarrow C_p > 0 \quad (2.74)$$

$$\text{If } p < p_\infty \rightarrow U > U_\infty \rightarrow C_p < 0 \quad (2.75)$$

Solving Potential Flows (TSec. 3.11 & 3.8.2)

To recap we have shown that if the flow is **potential i.e., steady, incompressible & irrotational (inviscid)** the continuity eq. reduces to the Laplace Eq. (Eq. (2.43) & (2.45)) $\rightarrow \nabla^2 \phi = \nabla^2 \psi = 0$.

- ▶ 1 equation, 1 unknown $\rightarrow$ Euler's eq. 4 eqns. 4 unknowns
- ▶ Linear equation $\rightarrow$ can use principle of superposition $\rightarrow$ if $\phi_1, \phi_2, \dots, \phi_n$ are all solutions to a linear equation then the linear combination of any of the solutions is also a solution. e.g., $\phi_i + 1/3\phi_j + 5\phi_k$ is also a solution. We shall exploit this. We shall find some basic solutions to the Laplace equation and then find linear combinations of them to yield other solutions.

Laplace Eq.

- ▶ Remember, $\nabla^2\phi = \nabla^2\psi = 0$, is an expression of the conservation of mass for **potential i.e., steady, incompressible & irrotational (inviscid)** flow. Also **2-d** in the case of ψ .
- ▶ So what happens to the momentum equations? → **Bernoulli's equation $p/\rho + U^2/2 = \text{const}$, everywhere** . We use $\nabla^2\phi = \nabla^2\psi = 0$ to solve for u and v and then use **Bernoulli's eq.** to calculate the pressure
- ▶ If all potential flows were governed by Laplace and Bernoulli's eqns. how do we distinguish between flows? for example, flow over an airfoil and that over a cylinder? → **boundary conditions**

Boundary conditions

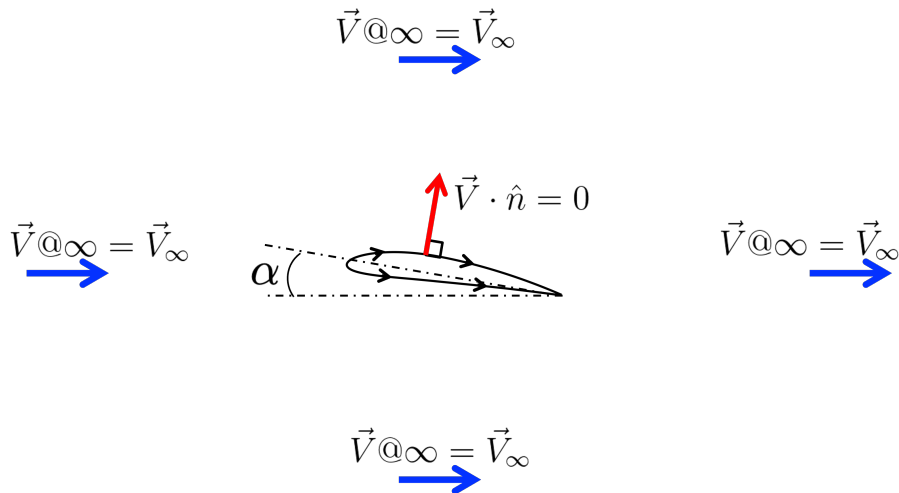


Figure 2.13: Boundary conditions for potential flow over an airfoil. Based on image from FA-JA

Boundary conditions

Far away from the body (∞),

$$\vec{V} = \vec{U}_{\infty} \quad (2.76)$$

For the airfoil shown this is expressed as,

$$u = \frac{\partial \phi}{\partial x} = \frac{\partial \psi}{\partial y} = U_{\infty} \cos \alpha \quad (2.77)$$

$$v = \frac{\partial \phi}{\partial y} = -\frac{\partial \psi}{\partial x} = U_{\infty} \sin \alpha \quad (2.78)$$

Boundary conditions

At the wall for potential flows we need to be careful. Consider the 2-D incompressible Navier-Stokes equations – continuity and momentum only. We have 3 equations (continuity, x -momentum and y -momentum) and 3 unknowns (p, u, v). So while solving these set of equations, on the wall of the airfoil shown above (and analogously any body in a real flow) we can enforce $p = p_{wall}, u = u_{wall} = 0$ & $v = v_{wall} = 0$. Note, $u = u_{wall} = 0$ & $v = v_{wall} = 0$ together is called the **no-slip boundary condition**. However, in potential flows we have only one equation to solve for both u and $w \rightarrow \nabla^2 \psi = 0$ or $\nabla^2 \phi = 0$. Hence, we can only impose one boundary condition at the wall.

We say that the flow cannot **penetrate** the surface of the airfoil or there is **no** flow through the airfoil surface

Boundary conditions

No flow through the surface i.e.,

$$\vec{V} \cdot \hat{n} = 0 \quad (2.79)$$

Note, for potential flow

$$\vec{V} = \vec{\nabla} \phi \rightarrow \vec{\nabla} \phi \cdot \hat{n} = 0 \quad (2.80)$$

In terms of the stream function

$$\frac{\partial \psi}{\partial s} = 0 \quad (2.81)$$

Therefore, on the surface of the body

$$\delta \psi = 0 \rightarrow \psi = \text{const} \quad (2.82)$$

Therefore, the surface of the body (airfoil) is a **streamline**.

Elementary flows (TSec. 3.12)

To recap -

- ▶ If the flow is steady and incompressible then by defining ψ we satisfy continuity automatically.
- ▶ If the flow is irrotational (inviscid) then $\vec{V} = \vec{\nabla}\phi$
- ▶ If the flow is steady, incompressible & irrotational (inviscid), the continuity equation reduces to Laplace equation $\nabla^2\phi = 0$.
- ▶ If the flow is steady, incompressible, & irrotational (inviscid) then ψ also satisfies the Laplace equation $\nabla^2\psi = 0$.
- ▶ Laplace equation is a linear equation. $\rightarrow$ can use **principle of superposition**

In this next section, we are going to establish some basic solutions or elementary flows. We are then going to use linear superposition to analyze flow over various objects including airfoils.

Uniform flow

$$\text{---} \phi = \text{const} \quad \text{---} \psi = \text{const}$$

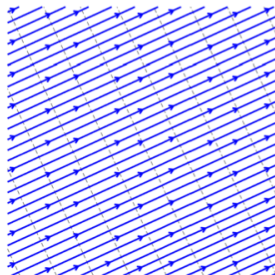


Figure 2.14: Uniform potential flow

$$u = U_{\infty} \cos \alpha \quad v = U_{\infty} \sin \alpha \quad (2.83)$$

$$\phi = U_{\infty} (x \cos \alpha + y \sin \alpha) \quad \psi = U_{\infty} (y \cos \alpha - x \sin \alpha) \quad (2.84)$$

$$\phi = U_{\infty} r (\cos \theta \cos \alpha + \sin \theta \sin \alpha) \quad \psi = U_{\infty} r (\sin \theta \cos \alpha - \cos \theta \sin \alpha) \quad (2.85)$$

Source/sink flow

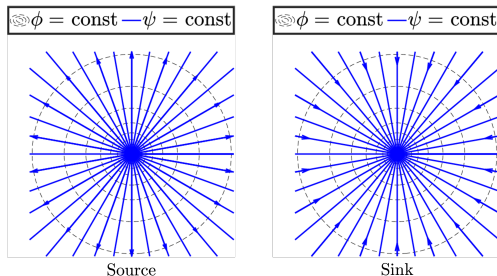


Figure 2.15: Source/sink flow

$$v_r = \frac{K}{2\pi r} \quad v_\theta = 0 \quad (2.86)$$

$$\phi = \frac{K}{2\pi} \ln r \quad \psi = \frac{K}{2\pi} \theta \quad (2.87)$$

$$K > 0 \text{ source} \quad K < 0 \text{ sink} \quad (2.88)$$

Note: θ is in radians, and the source and sink are located at the origin.

Derived flow - doublet flow

We are going to use the principle of superposition to derive our third flow - a doublet. Consider a source and a sink as shown below.

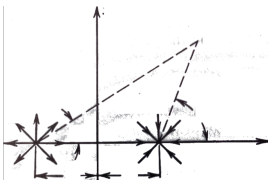


Figure 2.16: doublet potential flow, adapted from FA-KC

The stream function for the entire flow field is now

$$\psi = \psi_{\text{source}} + \psi_{\text{sink}} \quad (2.89)$$

$$= -\frac{K}{2\pi} (\theta_1 - \theta_2) = -\frac{K}{2\pi} \left(\tan^{-1} \frac{y}{x - x_0} - \tan^{-1} \frac{y}{x + x_0} \right) \quad (2.90)$$

Derived flow - doublet flow

$$\tan^{-1} \alpha - \tan^{-1} \beta = \tan^{-1} \left(\frac{\alpha - \beta}{1 + \alpha\beta} \right) \quad (2.91)$$

After some algebra, $\psi(x, y) = -\frac{K}{2\pi} \tan^{-1} \frac{2x_0 y}{x^2 + y^2 - x_0^2}$ (2.92)

let, $B = 2x_0 K \rightarrow$ doublet strength when $x_0 \rightarrow 0$ (2.93)

therefore, $\psi_{doublet} = \lim_{x_0 \rightarrow 0} \left[-\frac{B}{4\pi x_0} \tan^{-1} \left(\frac{2x_0 y}{x^2 + y^2 - x_0^2} \right) \right]$ (2.94)

$$x_0 \rightarrow 0, \frac{2x_0 y}{x^2 + y^2 - x_0^2} \rightarrow \text{something small} \quad (2.95)$$

therefore as $x_0 \rightarrow 0, \tan^{-1} \left(\frac{2x_0 y}{x^2 + y^2 - x_0^2} \right) \rightarrow \frac{2x_0 y}{x^2 + y^2 - x_0^2}$ (2.96)

Derived flow - doublet flow

$$\text{therefore, } \psi_{\text{doublet}} = \lim_{x_0 \rightarrow 0} \left[-\frac{B}{4\pi x_0} \frac{2x_0 y}{x^2 + y^2 - x_0^2} \right] \quad (2.97)$$

$$\rightarrow = -\frac{B}{2\pi} \frac{y}{x^2 + y^2} = -\frac{B}{2\pi} \frac{\sin \theta}{r} \quad (2.98)$$

$$v_r = -\frac{B}{2\pi} \frac{\cos \theta}{r^2} \quad v_\theta = -\frac{B}{2\pi} \frac{\sin \theta}{r^2} \quad (2.99)$$

$$\phi = \frac{B}{2\pi} \frac{\cos \theta}{r} \quad \psi = -\frac{B}{2\pi} \frac{\sin \theta}{r} \quad (2.100)$$

Note, Even though we derived this flow, now that we have the final result i.e., ψ and ϕ , the doublet also becomes an elementary flow. We can just use it, by adding it to other flows. This is the power of linear models!

Derived flow - doublet flow

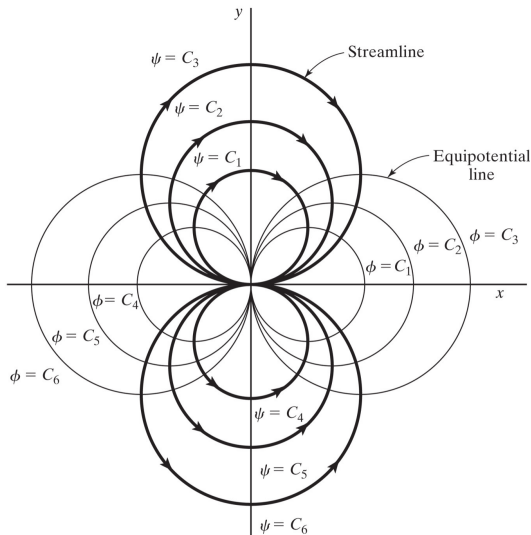


Figure 2.17: Fig. 3.11 from AE-BC

Vortex flow

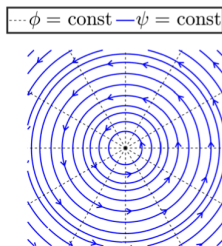


Figure 2.18: Vortex potential flow

$$v_r = 0 \quad v_\theta = -\frac{\Gamma}{2\pi r} \quad (2.101)$$

$$\phi = -\frac{\Gamma}{2\pi}\theta \quad \psi = \frac{\Gamma}{2\pi}\ln r \quad (2.102)$$

Note: θ is in radians, as in the source and the sink Γ is centered at the origin.

Summary

TABLE 3.3 Stream Functions and Potential Functions for Elementary Flows

Flow	ψ	ϕ
Uniform flow in x direction	$U_{\infty}r \sin \theta, U_{\infty}y$	$U_{\infty}r \cos \theta, U_{\infty}x$
Uniform flow at angle α	$U_{\infty}(y \cos \alpha - x \sin \alpha)$	$U_{\infty}(x \cos \alpha + y \sin \alpha)$
Source	$\frac{K\theta}{2\pi}$	$\frac{K}{2\pi} \ln r$
Doublet	$-\frac{B}{r} \sin \theta$	$\frac{B}{r} \cos \theta$
Vortex (with clockwise circulation)	$\frac{\Gamma}{2\pi} \ln r$	$-\frac{\Gamma\theta}{2\pi}$
90° corner flow	Axy	$\frac{1}{2}A(x^2 - y^2)$
Solid-body rotation	$\frac{1}{2}\omega r^2$	Does not exist since flow is rotational

Figure 2.19: Table 3.3 from AE-BC

Non-spinning cylinder (TSec. 3.13)

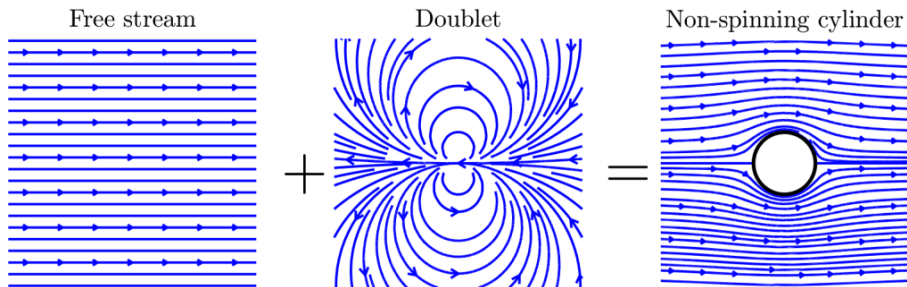


Figure 2.20: Potential flow over a non-spinning cylinder. Based on image from FA-JA

$$\psi(r, \theta) = \psi_{\text{free-stream}} + \psi_{\text{doublet}} \quad (2.103)$$

$$= U_{\infty} r \sin \theta + \frac{-B \sin \theta}{2\pi r} \quad (2.104)$$

$$= U_{\infty} r \sin \theta \left(1 - \frac{B}{2\pi U_{\infty} r^2} \right) \quad (2.105)$$

Potential flow over a non-spinning cylinder

$$\psi = U_{\infty} r \sin \theta \left(1 - \frac{B}{2\pi U_{\infty} r^2} \right) \quad (2.106)$$

$$v_r = \frac{1}{r} \frac{\partial \psi}{\partial \theta} = U_{\infty} \cos \theta \left[1 - \frac{B}{2\pi U_{\infty} r^2} \right] \quad (2.107)$$

$$v_{\theta} = -\frac{\partial \psi}{\partial r} = -U_{\infty} \sin \theta \left[1 + \frac{B}{2\pi U_{\infty} r^2} \right] \quad (2.108)$$

$$\text{Stagnation pt. } |\vec{V}| = 0 \rightarrow r = R = \sqrt{\frac{B}{2\pi U_{\infty}}} \quad (2.109)$$

$$C_p = 1 - \left(\frac{U}{U_{\infty}} \right)^2 = 1 - \left(\frac{\sqrt{v_r^2 + v_{\theta}^2}}{U_{\infty}} \right)^2 \quad (2.110)$$

$$C_p = 1 - 4 \sin^2 \theta \quad (2.111)$$

Potential flow over a non-spinning cylinder

$$\psi = U_{\infty} r \sin \theta \left[1 - \left(\frac{R}{r} \right)^2 \right] \quad (2.112)$$

$$\psi(r = R) = 0 \rightarrow \text{surface streamline} \quad (2.113)$$

Note, that it just happens that $\psi = 0$ is the surface streamline. There is no other significance to this.

- ▶ Stagnation point $\rightarrow \theta = 0 \text{ \& } \pi, r = R$. (Show as an ungraded homework)
- ▶ Max velocity $\rightarrow r = R, \theta = \pi/2 \text{ \& } 3\pi/2, v_r = 0, v_{\theta} = 2U_{\infty}$ (Show as an ungraded homework)
- ▶ At $r = R, v_r = 0$ and $v_{\theta} = -2U_{\infty} \sin \theta$ (Show as an ungraded homework)

Potential flow over a non-spinning cylinder

- ▶ $C_l = \frac{1}{c} \int_{le}^{te} (C_{pl} - C_{pu}) dy = -\frac{1}{2} \int_0^{2\pi} C_p \sin \theta d\theta = 0$. Can also be determined by observation of C_p distribution plot. There is top/bottom symmetry hence there is no net force in the y direction and thus no lift.
- ▶ $C_d = \frac{1}{c} \int_{le}^{te} (C_{pu} - C_{pl}) dx = \frac{1}{2} \int_0^{2\pi} C_p \cos \theta d\theta = 0$. Can also be determined by observation of C_p distribution plot. There is left/right symmetry hence there is not net force in the x direction and thus no drag.

$$C_d = \frac{1}{c} \int_{le}^{te} (C_{pu} - C_{pl}) dy \quad (2.114)$$

$$y = R \sin \theta \rightarrow dy = R \cos \theta d\theta, \quad c = 2R \quad (2.115)$$

$$C_d = \frac{1}{2} \int_0^{2\pi} C_p \cos \theta d\theta, \text{ Similarly } C_l = -\frac{1}{2} \int_0^{2\pi} C_p \sin \theta d\theta \quad (2.116)$$

Potential flow over a non-spinning cylinder

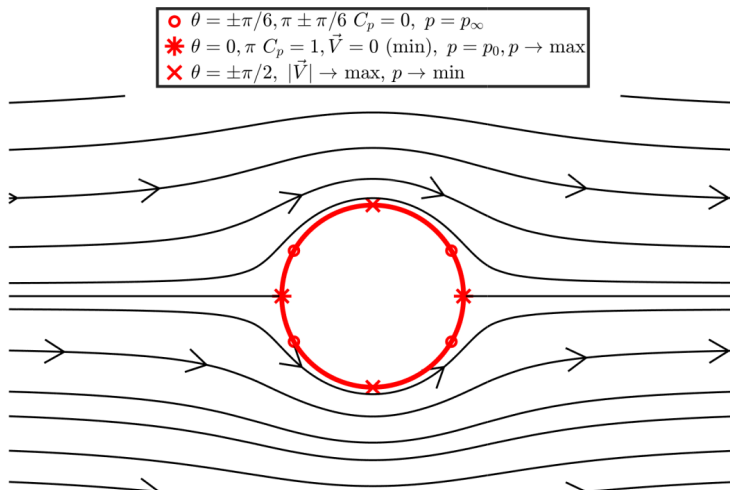


Figure 2.21: Potential flow over a non-spinning cylinder. Based on image from FA-JA

Potential flow over a non-spinning cylinder

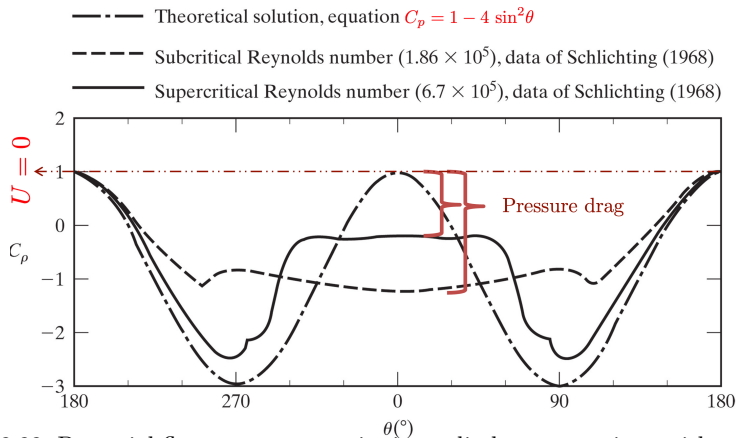


Figure 2.22: Potential flow over a non-spinning cylinder, comparison with experiments, modified Fig. 3.17 from AE-BC

Potential flow over a spinning cylinder (TSec. 3.15)

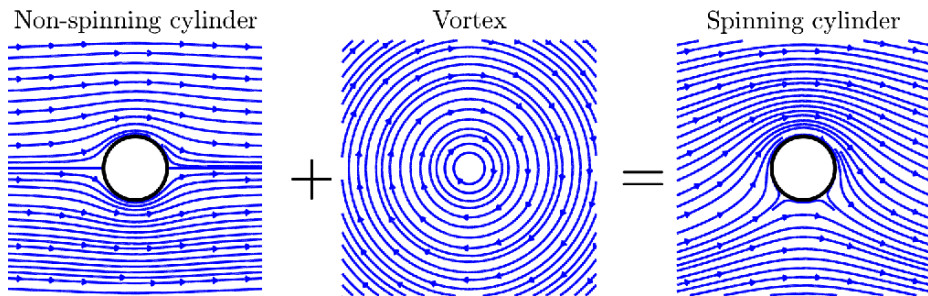


Figure 2.23: Potential flow over a spinning cylinder. Based on image from FA-JA

$$\psi(r, \theta) = \psi_{\text{free-stream}} + \psi_{\text{doublet}} + \psi_{\text{vortex}} \quad (2.117)$$

$$= \psi_{\text{non-spinning cylinder}} + \psi_{\text{vortex}} \quad (2.118)$$

Potential flow over a spinning cylinder

$$\psi(r, \theta) = U_{\infty} r \sin \theta \left[1 - \left(\frac{R}{r} \right)^2 \right] + \frac{\Gamma}{2\pi} \ln \frac{r}{R} \quad (2.119)$$

$$v_r = \frac{1}{r} \frac{\partial \psi}{\partial \theta} = U_{\infty} \cos \theta \left[1 - \frac{R^2}{r^2} \right] \rightarrow \text{same as non-lifting} \quad (2.120)$$

$$v_{\theta} = -\frac{\partial \psi}{\partial r} = \underbrace{-U_{\infty} \sin \theta \left[1 + \frac{R^2}{r^2} \right]}_{\text{non-lifting}} - \frac{\Gamma}{2\pi r} \quad (2.121)$$

$$C_p = \underbrace{1 - 4 \sin^2 \theta}_{\text{non-lifting}} + \frac{-2\Gamma \sin \theta}{\pi R U_{\infty}} - \left(\frac{\Gamma}{2\pi R U_{\infty}} \right)^2 \quad (2.122)$$

Potential flow over a spinning cylinder

$$C_d = \frac{1}{2} \int_0^{2\pi} C_p \cos \theta \, d\theta = 0 \quad (2.123)$$

$$C_l = -\frac{1}{2} \int_0^{2\pi} C_p \sin \theta \, d\theta = \frac{\Gamma}{RU_\infty} \text{ after some algebra} \quad (2.124)$$

$$l = \frac{1}{2} C_l \rho_\infty U_\infty^2 c = \frac{1}{2} \frac{\Gamma}{RU_\infty} \rho_\infty U_\infty^2 2R = \rho_\infty U_\infty \Gamma \quad (2.125)$$

$$C_d = 0, \quad C_l = \frac{\Gamma}{RU_\infty}, \quad l = \rho_\infty U_\infty \Gamma$$

“Spinning” the cylinder causes an asymmetrical force due to Γ which in turn causes lift $\rightarrow$ Magnus effect

Potential flow over a spinning cylinder

- ▶ Stagnation point $\rightarrow \theta = \sin^{-1} -\Gamma/4\pi RU_\infty$, $r = R$. (Show as an ungraded homework)
- ▶ Max velocity $\rightarrow r = R$, $\theta = \pi/2$ (Show as an ungraded homework)
- ▶ Stagnation streamline is still $\psi = 0$ (Show as an ungraded homework)
- ▶ At $r = R$, $v_r = 0$ and $v_\theta = -2U_\infty \sin\theta - \Gamma/2\pi R$ (Show as an ungraded homework)
- ▶ $C_l = \Gamma/RU_\infty$.
- ▶ $C_d = 0$. Can also determine by observation of C_p distribution plot. There is left/right symmetry hence there is not net force in the x direction and thus no drag.

Note: Since the spinning cylinder produces lift sometimes we refer to it as the lifting cylinder and the non-spinning cylinder as a non-lifting cylinder

Potential flow over a spinning cylinder

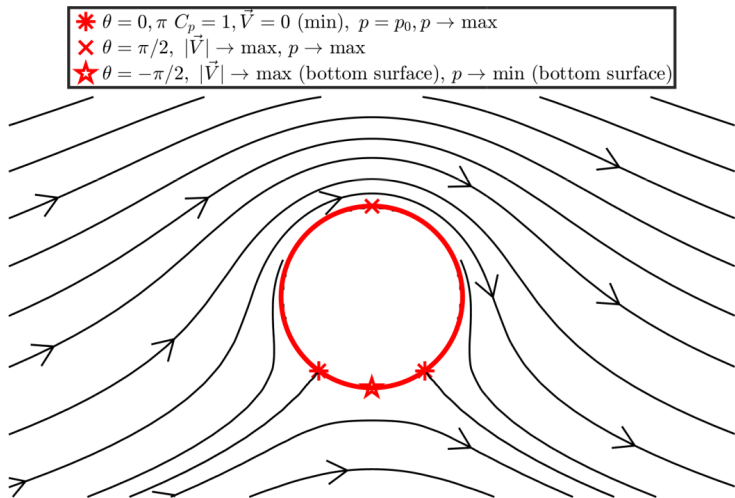


Figure 2.24: Potential flow over a spinning cylinder

Potential flow over a spinning cylinder

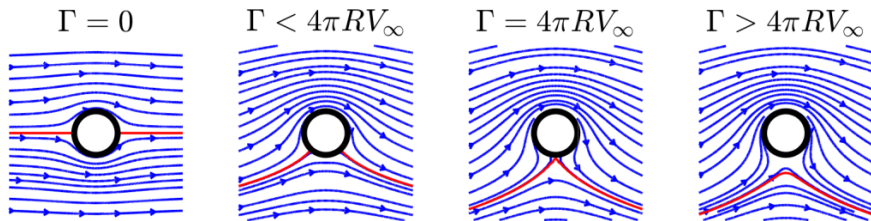


Figure 2.25: Potential flow over a spinning cylinder. Based on image from FA-JA

Γ determines the locations of the **stagnation** points. Γ is also directly proportional to how fast the cylinder is spinning and also to the lift.